AI 101 for Mechanical Engineers · Module 14

Retrieval supplies evidence at inference time

Lesson 14.2

What you will learn

> Explain retrieval-augmented generation and check whether a retrieved passage supports an engineering claim.

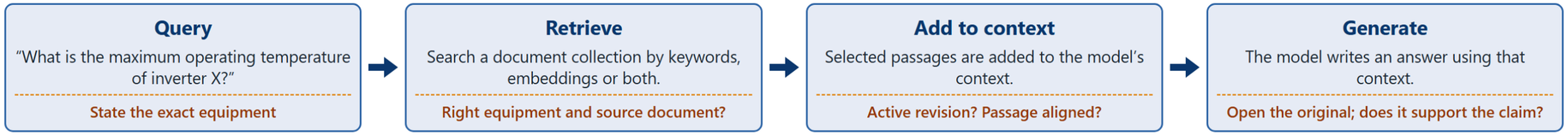
Lesson overview

Module 14 · Lesson 2

Retrieval supplies evidence at inference time

AI 101 for Mechanical Engineers

1 Retrieval-augmented generation (RAG), with a check at each stage



Supplying a passage at inference time does not update the model's weights. Training or fine-tuning a RAG system is a separate step.

2 Before using a retrieved operating limit

1. It applies to the target equipment and part number.
2. It comes from the active document revision.
3. Its actual wording, units and scope support the claim.

Hypothetical example:

✓ Inverter X manual, rev. C	✗ Inverter Y manual
Correct model and current revision	Looks relevant, but is invalid evidence for X

3 Words and their limits

Retrieval	Searching for material relevant to a query.
Grounding	Connecting an output to supplied evidence.
RAG	Retrieval, then adding the evidence to the context for generation.

Grounding makes review easier but does not remove misreading or unsupported claims. A citation label is not proof.

Record the source location so another engineer can repeat the check.

Key takeaway

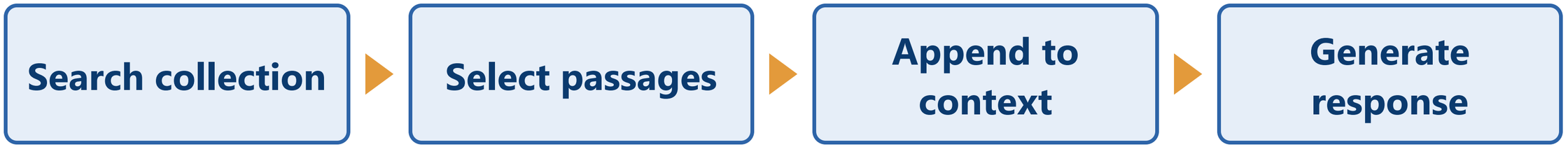
A retrieved passage is evidence to check, not an answer to trust.

Confirm equipment, revision and direct support in the original document.

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RAG adds retrieved evidence to the model context



- Searches for material relevant to a query [76]
- Retrieval may use keywords, embeddings or both

Supplying a passage is not retraining

At inference time

- Passage supplied in the context
- Model parameters are not updated

Training or fine-tuning

- A separate operation
- The original RAG study included fine-tuning [76]

Before relying on a retrieved passage

1. Confirm that the passage exists
2. Check it applies to the target equipment and part number
3. Check it belongs to the active revision
4. Check the actual wording supports the specific claim

Relevant-looking text can be invalid evidence

- A paragraph about a different inverter can look relevant
- It is still invalid evidence for the target inverter
- A citation label is not proof by itself
- Open the original: inspect passage, units, scope and revision

Grounding helps review but does not remove error

- Grounding connects a response to supplied evidence
- It can make review more direct
- Misreading, irrelevant retrieval and unsupported generation remain possible

Check the evidence at every stage

Stage	Check
Retrieve	Equipment identity and source document
Add context	Document revision and passage alignment
Generate and review	Original source directly supports the claim

> Preserve the source location so another engineer can reproduce the check.

Three terms describe information flow

Term	Meaning
Retrieval	Searching for material relevant to a query
Grounding	Connecting output to relevant supplied evidence
RAG	Retrieval, then evidence added to context for generation

> None of these terms guarantees that the answer is correct.

CHECK YOUR UNDERSTANDING

Does RAG retrain the model, and what must you check before using a retrieved operating limit?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does RAG retrain the model, and what must you check before using a retrieved operating limit?

> RAG does not inherently retrain model weights. Check the equipment and document revision, and that the passage actually supports the claim. Confident wording alone is not evidence.

Key takeaways

- Retrieval supplies context at inference time
- Check equipment identity, document revision and direct claim support
- Open original sources rather than trusting citation labels
- Grounding reduces review risk but does not remove model error

References

[76] P. Lewis et al., “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in Adv. Neural Inf. Process. Syst. (NeurIPS), vol. 33, 2020.

Numbers match the course reference list. The written lesson is the authoritative text.